

Venus Photochemistry

M.Tech. Project, Phase 1 Presentation



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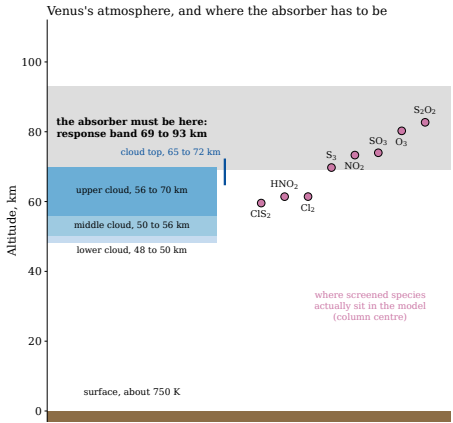
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Outline

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4. Experiments 8 to 14
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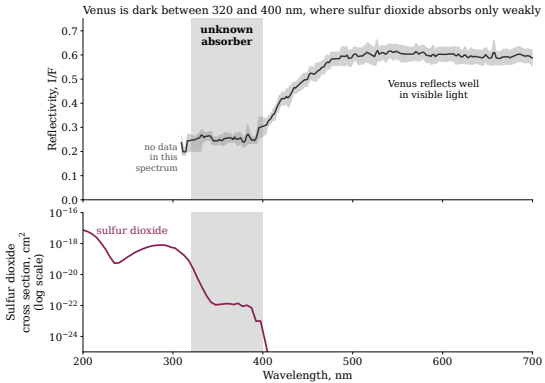
Venus and its clouds



Dai et al. (2025)

- Surface: 92 bar, about 750 K.
- Sulfuric acid clouds, 47 to about 70 km.
- Cloud top: 65 to 72 km.
- Grey band: where the model responds, 69 to 93 km.

The unknown ultraviolet absorber



Observed: MESSENGER spectrum, digitised from Pérez-Hoyos et al. (2018), Figure 8. Grey band: 320 to 400 nm.

Pérez-Hoyos et al. (2018); Dai et al. (2025); Titov et al. (2008)

- Dark at 320 to 400 nm; SO_2 cannot explain it.
- About half the solar heating at the cloud top.
- Amount varied twofold, 2006 to 2017.
- Its markings trace vertical motion.

The candidates proposed so far

Candidate	For	Against
Sulfur dioxide, SO_2	Matches below 320 nm	Negligible beyond 320 nm
Polysulfur, S_x	Early spectral candidate	No form matches
Ferric chloride, $FeCl_3$	Dissolves in the acid	Band too narrow and weak
Disulfur monoxide, S_2O	Matches 400 to 500 nm	Fades at 320 to 340 nm
OSSO, S_2O_2	Matches 330 to 420 nm	Far too little at cloud top
Iron and sulfur minerals	Absorb at 300 to 500 nm	Presence unknown
Chlorine and sulfur compounds	Spectra close	No abundance given
Organic “red oil”	Dissolves in acid	Speculative
A mixture of absorbers	Explains the band’s width	Not yet tested

Almost every case rests on the spectrum alone.

Dai et al. (2025); Pérez-Hoyos et al. (2018)

Literature survey: twelve papers, three tests

Paper	Focus	Spectrum	Amount	Altitude
Dai et al. (2024)	The model, OSSO	◔	◑	◔
Pérez-Hoyos et al. (2018)	Observed absorber	◑	○	◔
Bierson and Zhang (2020)	Mixing, OSSO	◔	◑	◑
Rimmer et al. (2021)	Droplet chemistry	◑	◔	◔
Lefèvre et al. (2022)	Cloud convection	○	◔	○
Lefèvre et al. (2024)	With chemistry	○	○	○
Jiang et al. (2024)	Iron sulfates	◑	◑	○
Trabelsi and Francisco (2026)	Chlorine and sulfur	◑	○	○
Egan et al. (2025)	OSSO, end to end	●	●	◑
Spacek et al. (2026)	Strength needed	◑	●	◔
Egan, James and Plane (2025)	$FeCl_3$, laboratory	◑	○	○
Egan et al. (2026)	$FeCl_3$, 3D model	●	◑	◔
This project	**All in one model**	◑	◑	●

Key

- ○ Not addressed
- ◔ Assumed or set
- ◑ Partly tested
- ● Tested in full

Partly: shape only, an estimate, or no light calculation.
Full: absolute, against observation.

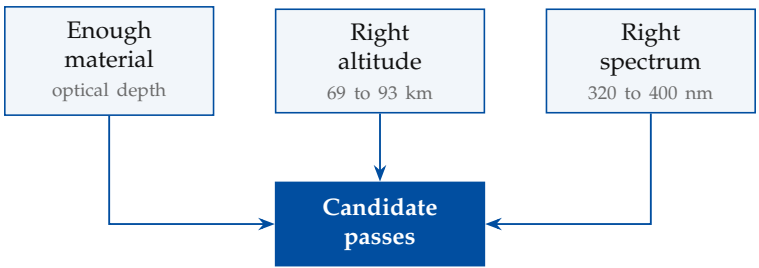
The twelve papers reviewed; five more references are background.

In the literature, altitude is never tested in full: it is assumed, set, or left out.

Research gap and objective

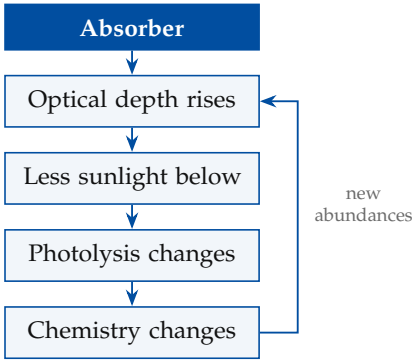
Gap: no study lets a model place a candidate and measures the effect of where it sits.

This project: all three tests at once, inside one model



The model: VULCAN-Venus

How an absorber acts on the model

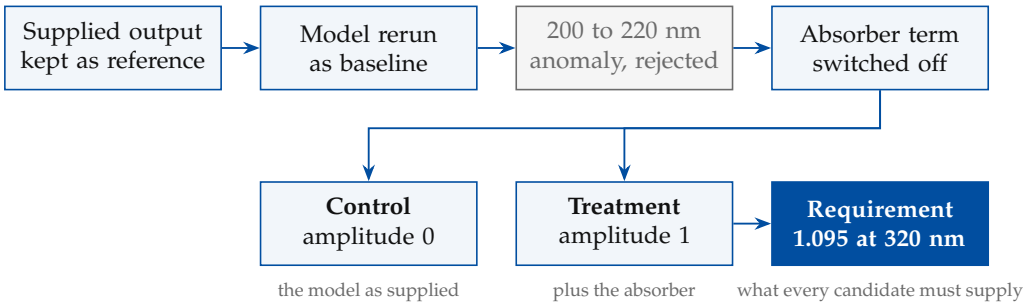


Dai, Shao and Sheng (2024); Bierson and Zhang (2020)

Set-up of every Phase 1 run

Grid	0 to 112 km, 57 layers
Chemistry	62 entries, 838 reaction slots
Wavelengths	1670 bins, 96 to 698 nm
Sunlight	One angle, 58 degrees
Mixing	Bierson and Zhang
Off	Vertical wind

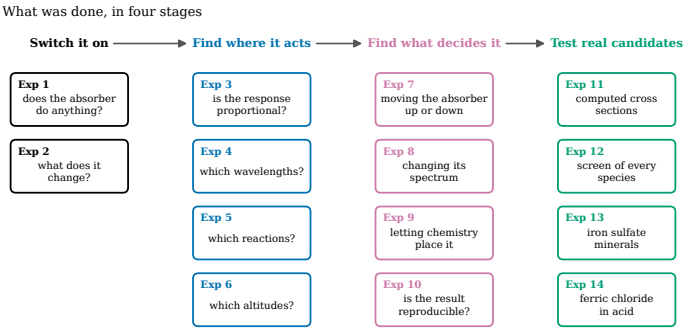
The groundwork: an absorber that was switched off



Control and treatment differ in one term only.

What was done, in four stages

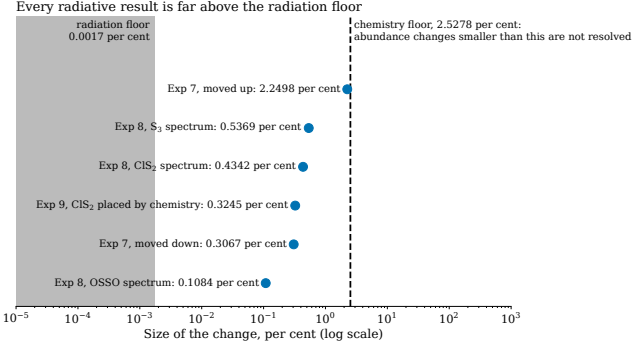
Fourteen experiments; each stage builds on the one before.



- Each effect is one number: the change in sunlight, in per cent of the sunlight itself.

How small a change is real

Experiment 10 repeats Experiment 9; the gap between them is numerical noise.



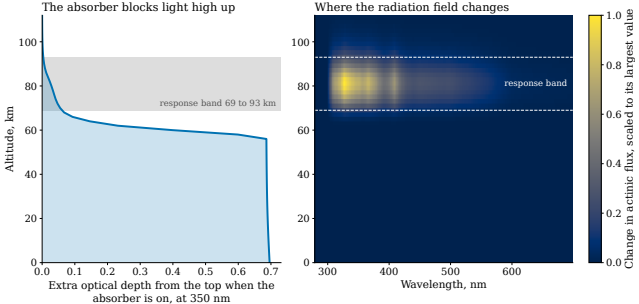
- Sunlight: the runs differ by 0.0017% of the change. Resolved.
- SO in the band: 0.7384% against 2.5278%. Noise.

The rule

Abundance changes below 2.5278% are upper bounds, not measurements.

Experiments 1 and 2: does the absorber do anything?

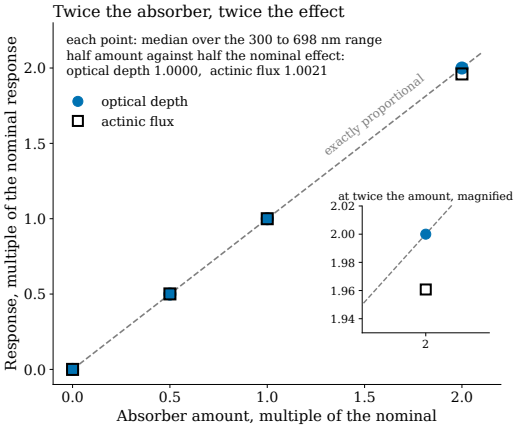
Switched on, the absorber changes sunlight strongly and the sulfur species barely.



	Largest change	Verdict
Sunlight	87.404%	resolved
SO_2	0.2243%	below floor
SO	1.7399%	below floor

Experiment 1: the control matches the supplied model, mean change 0.013553%.

Experiment 3: is the response proportional?

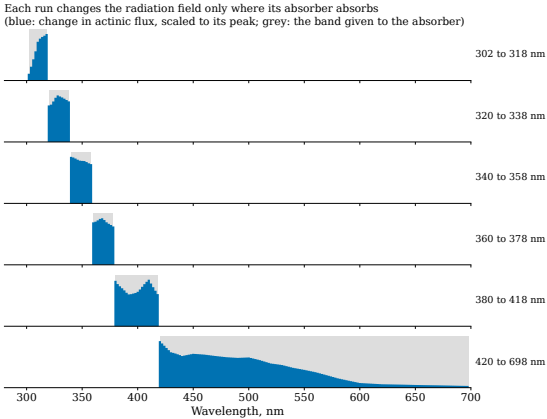


Half, nominal and twice the amount: the response scales with it.

Test, where 1 means proportional	Ratio
Optical depth, half amount	1.00002
Sunlight, half amount	1.00208
Whole field, half amount	1.010795
Whole field, twice the amount	0.979304

- Chemistry: within the floor, not resolved.

Experiment 4: which wavelengths?

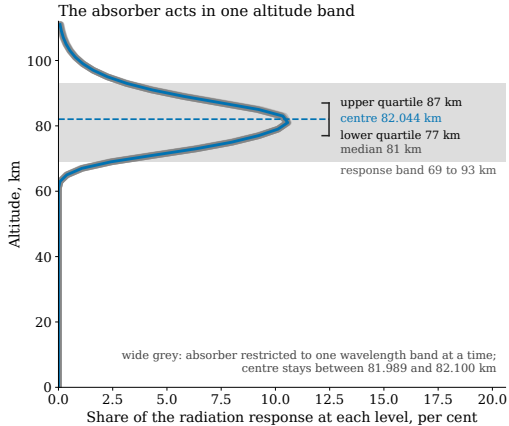


Each run limited to one window changes sunlight only in that window.

Molecule	Main window, nm	Share
S_2O	300 to 320	93.689%
SO_3	300 to 320	80.754%
S_3	380 to 420	49.605%
S_4	420 to 620	100%

- Each molecule draws on its own part of the spectrum.

Experiments 5 and 6: which reactions, which altitudes?



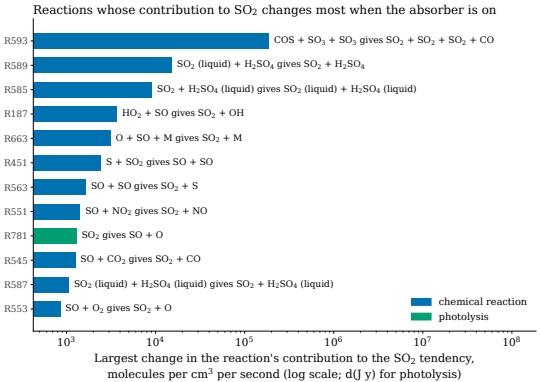
The response sits in one band, 69 to 93 km, for every wavelength window tested.

Reaction	In the band	Peak
SO_2 photolysis	99.128%	71 km
SO_3 photolysis	99.992%	73 km
COS + 2 SO_3	almost none	35 km

- Direct effects sit in the band; knock-on effects lie lower.

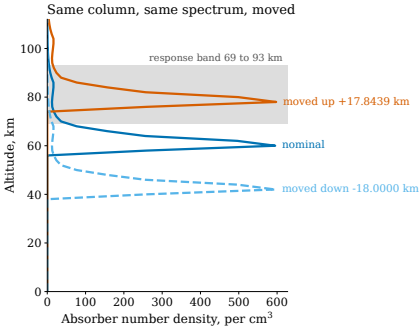
Experiment 5: the reactions behind the SO_2 change

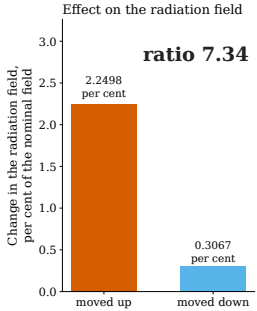
The largest absolute change is not the most affected reaction.



Experiment 7: does placement matter?

Moved up, the same absorber has 7.34 times more effect than moved down.



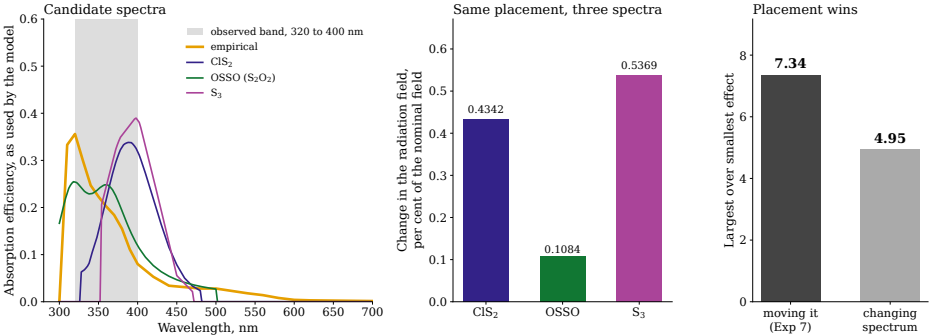


Move	Change
Up, +17.8439 km	2.2498%
Down, −18.0000 km	0.3067%
Ratio	**7.34**

- As supplied: 90.194% at 58 to 69 km, below the band.
- Moved up, SO in the band: 8.404%.

Experiment 8: spectrum against placement

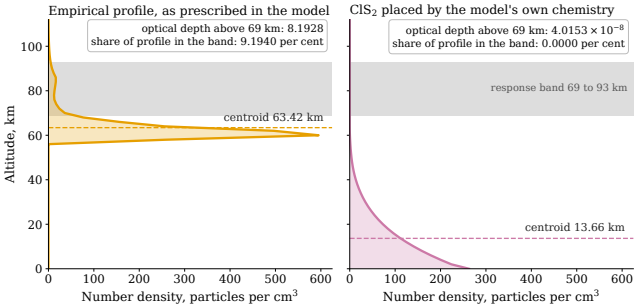
Same amount, same height, three spectra: shape matters, placement matters more.



- Spectra scaled to one total; optical depth above 69 km within 0.6461%.

Experiments 9 and 10: the chemistry places the candidate

Placed by its own chemistry, ClS_2 falls out of the response band.

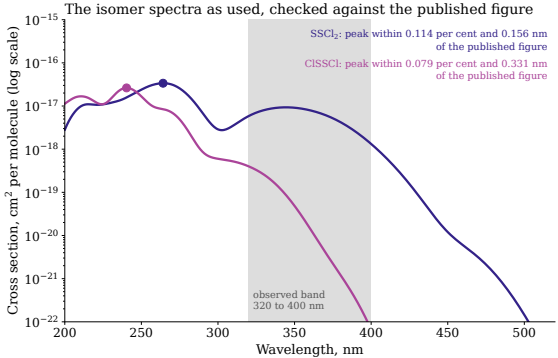


	Set	Chemistry
Centre, km	63.419	13.656
In band	9.1940%	0.0000%

- Closest in amount, short by 121.914.
- Above 69 km: 2.0×10^8 times less optical depth.
- Same as no absorber, to 0.001487%.
- Experiment 10, rerun with S_3: the same.

Experiment 11: two computed isomer spectra

The authors' cross sections run as absorbers; neither isomer can be made in the amount needed.



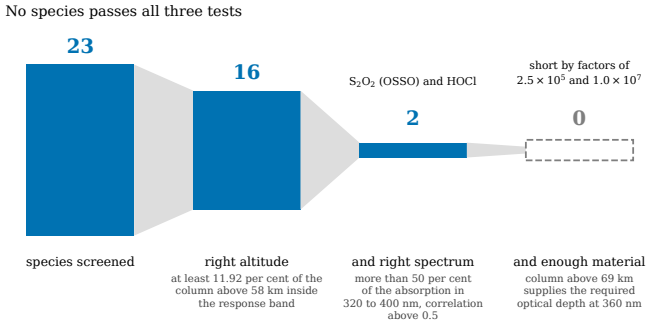
Trabelsi and Francisco (2026)

	$SSCl_2$	ClSSCl
Peak height	0.114%	0.079%
Peak position	0.156 nm	0.331 nm
Field change	0.19465%	0.30144%
Short by	4.266×10^5	6.668×10^6

- $SSCl_2$: 85.49% of its absorption at 320 to 400 nm, the most tested.
- ClSSCl peaks below 300 nm; the model sees its tail.

Experiment 12: all three tests on every species

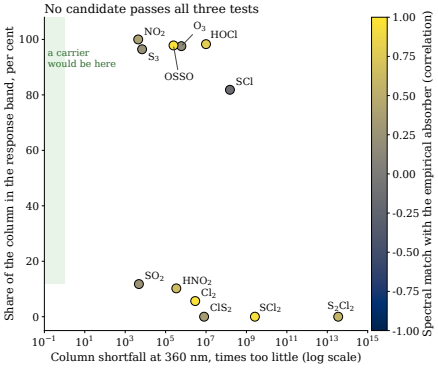
Twenty-three species screened; none passes all three tests.



Every pass mark is one the empirical absorber, which reproduces the observed brightness of Venus, itself meets.

Experiment 12: the three tests disagree

Amount is a gate; on the other two tests each picks a different best species.



Best eight on each test, best first

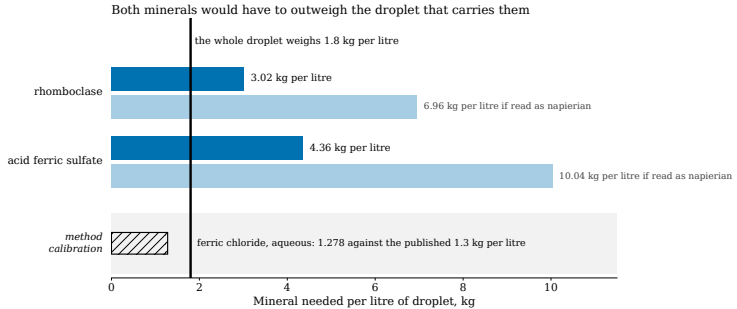
smallest shortfall	most column in the band	best spectral match
NO_2	NO_3	Cl_2
SO_2	NO_2	OSSO
S_3	$ClCO_3$	SCl_2
OSSO	SO_3	HOCl
HNO_2	ClO	HNO_2
O_3	S_2	S_2Cl_2
Cl_2	HOCl	H_2O_2
ClS_2	OSSO	NO_2

Each test picks a different winner: the rankings disagree.

no absorption at this wavelength, so not placed: $ClCO_3$, ClO, H_2O_2, H_2SO_4, HNO_3, NO_3, S_2, S_2O, S_4, SO_2Cl_2, SO_3

Experiment 13: iron and sulfur minerals

Even a droplet made wholly of either mineral could not absorb enough at 375 nm.

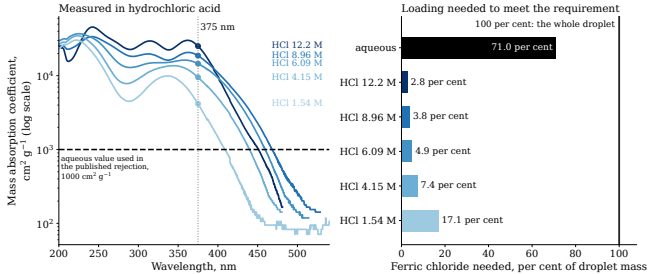


- Requirement: 1278 per cm at 375 nm; coefficients as measured by Jiang et al.

Jiang et al. (2024); Spacek et al. (2026)

Experiment 14: ferric chloride in acid

Measured in hydrochloric acid, ferric chloride needs 2.82% to 17.05% of the droplet, not 71.0%.



Most chloride-rich curve: a factor of 2.6 from the independent Egan et al. (2026) fit, against 64.5 before the correction

- The rejection rested on water; altitude is still untested.

Spacek et al. (2026); Egan, James and Plane (2025); Egan et al. (2026)

What Phase 1 establishes

Result	Basis
The absorber acts within one band, 69 to 93 km	Measured
Placement beats spectrum: 7.34 against 4.95	Measured
Placed by its own chemistry, ClS_2 equals no absorber	Measured
No species in the model passes all three tests	Calculated, model output
Minerals fall short; the ferric chloride rejection is reopened	Calculated, published data
Chemical responses mostly below the floor	Bounded

- Measured: radiative, reproducible to 0.0017% of the change.
- Bounded: abundances, reproducible only to 2.5278%.

The limits

Limit	What it means
One column, one sun angle	No latitude, no local time, steady states only
Weak vertical mixing	Placement tested at one mixing strength
Simplified droplets	Fitted cross sections; full droplet calculation off
Empirical reference	Spectral rankings measure resemblance to a fit
Only what the network holds	23 of 62 species could be screened
Floor from one pair of runs	A lower bound on the numerical scatter

Every run shares these limits, so comparisons stand; Phase 2 takes up the first two.

Verification: no number typed by hand

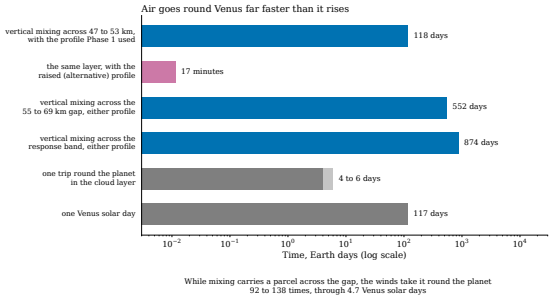
Every number is recomputed from the saved output and checked on the page.



- The checker proves a number is in the registry, not that it is the right one.
- So every number was also checked by hand against its own entry.

Future work

Phase 2: can realistic transport carry a candidate into the response band?



Mixing needs 552 days to cross the 55 to 69 km gap; the winds circle Venus in 4 to 6.

Limaye et al. (2018); Bertaux et al. (2016)

Six work packages

WP1 Stronger mixing	WP2 Cloud-top layer
WP3 Upwelling	WP4 Noon column
WP5 Lifetime test	WP6 Ferric chloride

WP1 and WP5 first, ferric chloride last.

Summary

- Switched on, the absorber acts in one band: 69 to 93 km.
- Where an absorber sits matters more than its spectrum.
- The chemistry, not the modeller, decides where a candidate sits.
- No species in the model passes all three tests.
- Minerals fall short; ferric chloride is reopened, its altitude untested.
- Every number traces back to the saved output.

Next: can realistic mixing carry a candidate into the band?

References

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Trabelsi, T., and Francisco, J. S. (2026). Chlorine-sulfur isomers as parents of ClS_2 and SCl_2 on Venus: spectroscopy and photochemistry of ClSSCl, $SSCl_2$, and $(ClS)_2$. *J. Chem. Phys.* 164, 084302.

Numbers and their sources (1 of 4)

Number	What it is	Registry entry	Report
1.095	Optical depth needed at 320 nm	`const.tau_req`	3.6
2.2498%	Field change, absorber moved up	`e7a.L2pct`	5.6
0.3067%	Field change, absorber moved down	`e7b.L2pct`	5.6
7.34	Ratio of the two moves	`e7.ratio_7A_over_7B`	5.6
69 to 93 km	Response band, 90 per cent share	`e6.interval.90.lo, .hi`	4.5
90.194%	Absorber at 58 to 69 km, as supplied	`pnom.cloud`	4.5
0.0017%	Radiation floor, share of the change	`e10.fielddiff_pct`	6.3
0.7384, 2.5278%	SO in the band, Experiments 9 and 10	`res.e9.SO.inband, res.e10.SO.inband`	6.3
0.013553%	Experiment 1, mean change	`e1.aflux.meanrel`	5.1

Set-up values come from the model files (report Table 3.1); literature values are cited on their slides.

Numbers and their sources (2 of 4)

Number	What it is	Registry entry	Report
87.404%	Largest change in sunlight	`e3fp.bnd.A1.aflux.max_pct`	5.1
0.2243, 1.7399%	Largest SO_2 and SO changes	`e2.SO2.maxrel, e2.SO.maxrel`	5.1
1.00002, 1.00208	Proportionality, optical depth and flux	`e3.lin.tau.inband.median, e3.lin.aflux.inband.median`	5.2
1.010795, 0.979304	Proportionality at half and twice the amount	`e3fp.aflux.ratio.A0p5, .A2p0`	5.2
93.689, 80.754, 49.605, 100%	Main-window shares, S_2O, SO_3, S_3, S_4	`e4a.<species>_1.<window>.pct`	5.3
99.128%, 71 km	SO_2 photolysis in the band, its peak	`e6.R781.pct_in_band, .peak_z`	5.5
99.992%, 73 km	SO_3 photolysis in the band, its peak	`e6.R787.pct_in_band, .peak_z`	5.5
35 km	Peak of the knock-on reaction	`e6.R593.peak_z`	5.5
17.8439, 18.0000 km	Centre moved up, moved down	`e7.7a.centroid_shift, e7.7b.centroid_shift`	5.6
8.404%	SO in the band, absorber moved up	`res.e7a.SO.inband`	5.6
0.6461%	Optical depth difference between spectra	`tau.spec_change_pct`	6.1
4.95	Largest over smallest spectral effect	`e8c.L2pct over e8b.L2pct`	6.1

Numbers and their sources (3 of 4)

Number	What it is	Registry entry	Report
121.914	ClS_2 shortfall, Experiment 8	`scr.ClS2.short`	6.1
63.419, 13.656 km	Centre, set and by chemistry	`pnom.centroid, p9.centroid`	6.2
9.1940, 0.0000%	Share in the band, set and by chemistry	`pnom.inband, p9.inband`	6.2
2.0×10^8	Fall in optical depth above 69 km	`tau.reduction`	6.2
0.001487%	Difference from no absorber	`id.l2_ratio_pct`	6.2
0.114%, 0.156 nm	$SSCl_2$ peak match	`iso.SSCl2.val_pct, .val_nm`	6.4
0.079%, 0.331 nm	ClSSCl peak match	`iso.ClSSCl.val_pct, .val_nm`	6.4
0.19465, 0.30144%	Field change, $SSCl_2$ and ClSSCl	`e11a.L2pct, e11b.L2pct`	6.4
4.266×10^5, 6.668×10^6	Isomer shortfalls	`iso.SSCl2.short, iso.ClSSCl.short`	6.4
85.49%	$SSCl_2$ share at 320 to 400 nm	`share320400.11A`	6.4

Numbers and their sources (4 of 4)

Number	What it is	Registry entry	Report
23, 16, 2, 0	Screened, placement, both, all three	`e12.n_screened`, `e12.gate.n_placement`, `.n_place_and_spec`, `.n_all`	6.5
62	Species the model tracks	`e12.n_tracked`	6.5
1278 per cm	Absorption the droplets need at 375 nm	`e13.A_required`	6.6
71.0%	Ferric chloride need, aqueous	`e14.aqueous_pct_of_droplet`	6.7
2.82, 17.05%	Ferric chloride need, hydrochloric acid	`e14.ClA.pct_of_droplet`, `e14.ClE.pct_of_droplet`	6.7
552 days	Mixing across the 55 to 69 km gap	`p2.gap.Bkzz.t_days`	9.1
10830	Values in the registry	all entries of `verify/registry.json`	8.1
407, 129	Numbers checked, report and figures	checker output	8.2

Thank you